HYBRID AI-BASED DETECTION OF FILELESS MALWARE IN IoMT HEALTHCARE SYSTEMS

INDIVIDUAL THESIS

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Component 03: Predictive IoMT Threat Forecasting Engine

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

This individual thesis presents the design, implementation and operational validation of a predictive security monitoring engine for Internet of Medical Things (IoMT) environments. As Component 3 of MediRecon, the engine integrates endpoint, process, PowerShell, authentication, network and DICOM-related telemetry into a fixed 28-feature representation. Five-minute observations are arranged into twelve-step sequences for an LSTM-based classification of next-24-hour security risk into LOW, MEDIUM, HIGH and CRITICAL categories. The implementation includes a central FastAPI service, distributed collectors, persistent SQLite histories, a dashboard, evidence-aware potential threat-spread analysis, and SOC notification with automatically attached PDF reports. Supplied system screenshots demonstrate operational telemetry, completed sequence history, risk trend visualization, risk-dependent response guidance and receipt of a CRITICAL email notification. The thesis distinguishes prospective risk estimation from confirmed incident detection and describes the interfaces needed for integration with the other MediRecon components.

Keywords: IoMT; cybersecurity; LSTM; threat forecasting; time-series telemetry; SOC

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1 INTRODUCTION

Chapter focus. Healthcare networks combine imaging services, clinical workstations, and distributed medical systems. MediRecon provides a research-oriented framework that combines DICOM network anomaly detection, endpoint fileless-malware analysis, future threat forecasting, and cross-layer correlation. The third component is a predictive decision-support capability rather than a direct malware classifier. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

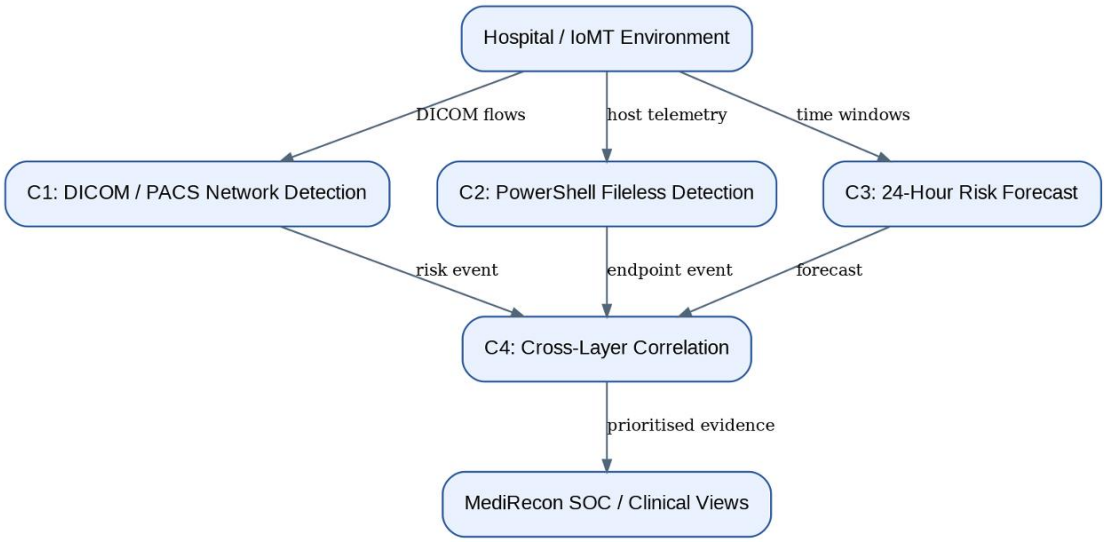


Figure 1.0.1. MediRecon overall research component architecture

1.1 Overall project context

The rapid adoption of Internet of Medical Things (IoMT) technologies has transformed healthcare environments by enabling interconnected medical devices, clinical workstations, diagnostic imaging systems, and network-based patient care services. These technologies support the efficient exchange of clinical information, improve access to medical resources, and contribute to the continuity of healthcare operations. However, the growing interconnectivity of healthcare infrastructure also introduces cybersecurity

challenges, particularly when medical systems operate with heterogeneous hardware, software, and communication protocols.

Fileless malware represents a significant concern in such environments because malicious activity may be performed through legitimate operating-system utilities, scripting environments, and administrative interfaces without relying on conventional executable files. Consequently, security monitoring approaches that depend primarily on file signatures may not provide sufficient visibility into suspicious host and network behavior.

The main research project, Hybrid AI-based Detection of Fileless Malware in IoMT Healthcare Systems, introduces MediRecon, a security monitoring framework designed to support the identification, prediction, and investigation of cybersecurity threats affecting healthcare infrastructure. The framework comprises four complementary research components: DICOM medical imaging network anomaly detection, PowerShell-based fileless malware detection, predictive IoMT threat forecasting, and cross-layer security correlation with explainable analysis.

Within this architecture, Component 3 provides a forward-looking cybersecurity capability. Rather than identifying only malicious events that have already occurred, it examines patterns in operational telemetry to estimate the level of cybersecurity risk that may arise during the following 24 hours. This enables security analysts to consider emerging risk conditions alongside current alerts and observed activity.

The integration of predictive analysis with network monitoring, endpoint telemetry, and security reporting establishes a foundation for proactive security decision support in healthcare environments. Such an approach is particularly relevant to IoMT systems, where security incidents may affect the availability of clinical services and the reliability of interconnected medical operations.

1.2 Individual component context

The individual research component addressed in this thesis is the Predictive IoMT Threat Forecasting Engine, designated as Component 3 of the MediRecon framework. Its primary

purpose is to analyze time-dependent security telemetry and produce a prediction of the cybersecurity risk level for the next 24 hours.

The component collects information from network communications, endpoint activities, PowerShell execution records, process events, authentication activity, and DICOM-related telemetry. These observations are transformed into a structured representation consisting of 28 numerical features. The feature set includes measurements such as network-flow counts, TCP connection indicators, SMB activity, PowerShell execution characteristics, process creation, authentication events, and medical imaging communication statistics.

To preserve temporal relationships, the system organizes the collected information into consecutive five-minute observation windows. A sequence of twelve windows represents one hour of operational activity. This sequence is processed by a Long Short-Term Memory (LSTM) neural network, which generates a classification for the subsequent 24-hour risk horizon.

The forecasting model produces four possible risk categories: LOW, MEDIUM, HIGH, and CRITICAL. These categories are presented through an interactive dashboard that includes a risk gauge, forecast history, current telemetry values, recommended security responses, and supporting investigation information.

The component also maintains persistent records of security observations and forecasting results. A supplementary evidence-correlation mechanism examines relationships between suspicious endpoint activity and relevant network communications to identify potential threat-propagation patterns. This mechanism supports investigation but remains separate from the LSTM forecasting process.

For operational reporting, the system provides security reports and SOC/Admin email notifications. The implemented notification workflow supports automatic delivery of a security report when a HIGH or CRITICAL forecast is generated. Accordingly, the component combines temporal prediction, security monitoring, evidence interpretation, and analyst-oriented reporting within a single forecasting service.

1.3 Problem statement

Healthcare security monitoring systems generate substantial quantities of operational information from network devices, clinical workstations, medical imaging services, and endpoint security mechanisms. Although these records can reveal individual suspicious events, interpreting them collectively to understand the development of future cybersecurity risk remains challenging.

Traditional monitoring approaches frequently emphasize the detection of current or previously observed security events. For example, a monitoring rule may identify an unusual PowerShell command, an abnormal network connection, or repeated authentication failures. While these observations are valuable, isolated alerts do not necessarily describe how the overall security condition of an IoMT environment may evolve over time.

This limitation becomes particularly important when suspicious activity develops gradually across multiple observation periods. Changes in network communication patterns, endpoint execution behavior, and authentication activity may collectively indicate an increasing level of security concern even when individual events require further investigation.

Another challenge involves the heterogeneous nature of healthcare telemetry. Network-flow measurements, PowerShell events, process information, authentication records, and DICOM communication statistics have different structures and meanings. A forecasting mechanism must transform these observations into a consistent representation while preserving their chronological relationships.

Therefore, the central research problem addressed by Component 3 is:

How can heterogeneous network, endpoint, authentication, and DICOM-related telemetry from healthcare IoMT environments be transformed into temporally structured information to predict cybersecurity risk levels for the next 24 hours and support proactive security decision-making?

Addressing this problem requires a consistent telemetry collection mechanism, time-based feature aggregation, sequential machine-learning inference, persistent forecast storage, and a means of presenting predictions together with relevant security evidence.

1.4 Research gap

Cybersecurity monitoring commonly relies on signature-based detection, predefined correlation rules, anomaly detection, and event-driven alerting. These approaches are important for identifying observable security conditions; however, detecting a current anomaly and forecasting future risk are distinct analytical objectives.

A current-event detection mechanism evaluates activity that has already been observed. In contrast, a forecasting mechanism examines historical sequences to estimate a future condition. The latter requires explicit consideration of observation intervals, temporal ordering, sequence construction, and the relationship between historical telemetry and the prediction horizon.

Within the context of this research, the identified gap concerns the need for an integrated forecasting capability that can transform heterogeneous IoMT security telemetry into a consistent time-series representation and present next-24-hour risk classifications in a form suitable for security operations.

The component addresses this gap through a fixed 28-feature telemetry structure, five-minute aggregation intervals, and twelve-window sequential processing using an LSTM model. This design allows the forecasting service to evaluate changes across an observation period rather than treating each security event as an independent prediction input.

An additional engineering requirement is the connection between predictive output and operational evidence. A risk classification alone does not identify the underlying cause of an incident or establish that a system has been compromised. Therefore, the proposed component presents predictions alongside telemetry indicators, persistent event records, security-response recommendations, and potential threat-propagation relationships.

The research contribution lies in combining these functions into an operational forecasting architecture for healthcare IoMT security monitoring, with particular emphasis on temporal consistency, interpretable security context, and integration with the wider MediRecon framework.

1.5 Aim and objectives

The primary aim of this research is to design and implement an AI-driven predictive cybersecurity engine capable of analyzing sequential IoMT security telemetry and forecasting the threat-risk level for the next 24 hours to support proactive security monitoring and incident-response planning in healthcare environments.

To achieve this aim, the following research objectives are established:

1. To identify and organize relevant network, endpoint, authentication, PowerShell, and DICOM-related telemetry into a consistent 28-feature input representation suitable for predictive analysis.

2. To develop a time-based telemetry processing mechanism that aggregates security observations into consecutive five-minute windows while maintaining chronological consistency.

3. To implement an LSTM-based forecasting mechanism that processes twelve consecutive telemetry windows and produces next-24-hour cybersecurity risk classifications across four categories: LOW, MEDIUM, HIGH, and CRITICAL.

4. To develop a centralized API and persistent storage mechanism for receiving security events, maintaining forecasting history, and supporting communication with other MediRecon components.

5. To design an interactive security dashboard that presents current predictions, historical risk trends, telemetry indicators, and recommended security responses.

6. To incorporate supporting evidence correlation and automated SOC/Admin reporting capabilities that assist analysts in investigating elevated cybersecurity risk.

These objectives define the technical and operational scope of the individual component and provide a basis for describing its design, implementation, and verification in subsequent chapters.

1.6 Research questions

The research is guided by a primary question and a set of supporting technical questions.

The primary research question is:

How can sequential machine-learning analysis of heterogeneous IoMT security telemetry support the prediction and operational interpretation of cybersecurity risk over a 24-hour forecasting horizon?

The supporting research questions are:

RQ1: How can network, endpoint, PowerShell, authentication, and DICOM-related security observations be normalized into a consistent feature structure for sequential analysis?

RQ2: How can fixed five-minute aggregation intervals and twelve-window sequences be used to preserve temporal relationships during LSTM-based cybersecurity forecasting?

RQ3: How can an LSTM forecasting service classify next-24-hour IoMT cybersecurity risk into LOW, MEDIUM, HIGH, and CRITICAL categories?

RQ4: How can forecasting results be integrated with persistent event records, security dashboards, and recommended response information to support SOC investigation?

RQ5: How can automated security reporting and evidence-based correlation provide additional operational context for elevated risk predictions without treating predictive classifications as confirmation of malicious compromise?

These questions guide the system design and establish the relationship between the forecasting model, telemetry infrastructure, and security operations functions.

1.7 Scope and Assumptions

The scope of this individual research is limited to the design, implementation, and operational evaluation of Component 3, the Predictive IoMT Threat Forecasting Engine.

The component covers the collection and ingestion of supported security telemetry, five-minute feature aggregation, sequential LSTM inference, next-24-hour risk classification, persistent forecasting history, dashboard visualization, security-response recommendations, and automated reporting. It also includes an evidence-correlation capability for presenting potential relationships between suspicious endpoint behavior and relevant network communications.

The forecasting model operates using a fixed input structure of 28 numerical features across twelve consecutive observation windows. This represents one hour of historical telemetry and supports a prediction horizon of the following 24 hours. The four output categories are LOW, MEDIUM, HIGH, and CRITICAL.

The component is designed to operate within a centralized monitoring architecture while receiving observations from distributed endpoint and network monitoring agents. It supports device identification and event association without requiring monitored device identities to be permanently embedded in the forecasting logic.

The wider MediRecon framework includes three other research components. Component 1 addresses DICOM medical imaging network anomaly detection, Component 2 focuses on PowerShell-based fileless malware detection, and Component 4 addresses cross-layer security correlation and explainable analysis. Although Component 3 provides interfaces for integration with these modules, their independent detection models and research contributions are outside the scope of this thesis.

The system assumes that incoming telemetry is timestamped, mapped to the required feature definitions, and processed in the expected chronological order. It also assumes that security analysts interpret forecasts together with available operational evidence. A HIGH or CRITICAL risk prediction indicates a need for prioritized investigation rather than independently proving that malware infection or lateral movement has occurred.

1.8 Individual Contribution

The principal individual contribution of this research is the development of an integrated, LSTM-based threat forecasting capability for healthcare IoMT security monitoring.

The component introduces a structured processing pipeline that converts heterogeneous security observations into five-minute feature windows and organizes twelve consecutive windows for sequential inference. This establishes a consistent relationship between the monitoring interval, historical context, and next-24-hour forecasting horizon.

A further contribution is the integration of forecasting with a centralized security monitoring service. The implementation includes event ingestion interfaces, dynamic device identification, distributed telemetry collection, persistent forecast records, and an interactive dashboard that communicates the current risk classification and historical trends.

The dashboard presents risk information through visual indicators and provides supporting context from network and endpoint telemetry. Security analysts can examine observed indicators, review recommended responses, and access generated security reports.

The component additionally implements automatic SOC/Admin notification for elevated forecasts. When the system produces a HIGH or CRITICAL risk classification, the notification workflow supports generating a security report and sending it to a configured recipient through email. This connects predictive output with an operational escalation mechanism.

An evidence-based potential threat-propagation capability is also included. It examines temporal relationships between suspicious endpoint observations and selected network communications, distinguishing correlated security evidence from ordinary network relationships. The resulting visualization assists investigation without presenting potential propagation as confirmed infection.

Collectively, these contributions establish Component 3 as a predictive and analyst-oriented cybersecurity service within MediRecon, complementing the detection and correlation functions of the other research components.

1.9 Thesis Structure

This thesis is organized into nine chapters, progressing from the research background and problem definition to the technical implementation and evaluation of the Predictive IoMT Threat Forecasting Engine.

Chapter 1 — Introduction presents the overall research context, the individual component, the problem statement, research gap, objectives, research questions, scope, and individual contribution.

Chapter 2 — Literature Review examines the relevant background in IoMT cybersecurity, fileless malware behavior, security telemetry analysis, time-series forecasting, and LSTM-based predictive techniques.

Chapter 3 — Research Methodology describes the approach used to structure security telemetry, define the forecasting problem, prepare sequential inputs, and establish the implementation and evaluation procedure.

Chapter 4 — System Design and Architecture explains the component architecture, telemetry collection mechanisms, data flow, forecasting pipeline, database organization, API interactions, and security integration interfaces.

Chapter 5 — System Implementation presents the development of the LSTM forecasting service, telemetry processing modules, persistent storage, dashboard, evidence-correlation mechanism, and automated reporting workflow.

Chapter 6 — Testing and Results documents the observed behavior of the implemented system, including telemetry processing, prediction presentation, event persistence, security monitoring, reporting, and integration testing supported by available evidence.

Chapter 7 — Discussion interprets the system's technical and operational outcomes in relation to the research objectives, forecasting design, healthcare monitoring requirements, and SOC decision-support needs.

Chapter 8 — Sustainability and Future Development discusses maintainability, deployment considerations, extensibility, and possible future enhancements to the forecasting service.

Chapter 9 — Conclusion summarizes the research contribution, the implemented forecasting capability, and its role within the MediRecon security framework.

Supporting architecture diagrams, telemetry illustrations, dashboard screenshots, and reporting evidence are incorporated into the relevant chapters to explain the implementation and substantiate the reported functionality.

2 LITERATURE REVIEW AND RELATED WORK

Chapter focus. Relevant literature concerns IoMT security, anomaly detection, multivariate time series, LSTM networks, SOC operations, and explainable incident investigation. This review distinguishes prediction of a future state from detection of a present attack. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

2.1 Review Method and Scope

The literature review provides the theoretical background for the development of the Predictive IoMT Threat Forecasting Engine. It focuses on cybersecurity challenges in healthcare environments and the use of Artificial Intelligence (AI) to support security monitoring and future threat prediction.

The review mainly covers IoMT security, fileless malware, machine learning, deep learning, Long Short-Term Memory (LSTM) networks, and Security Operations Center (SOC) operations. These areas are important because the proposed system collects

security information from different sources and uses it to predict possible cybersecurity risks.

Traditional cybersecurity systems mainly focus on detecting suspicious activities that have already occurred. However, predictive security systems attempt to identify future risk conditions by analyzing previous security activities. This difference is important for the proposed research because Component 3 focuses on predicting the cybersecurity risk level for the next 24 hours.

The review also examines how network monitoring, endpoint monitoring, and medical imaging traffic can provide useful information for security prediction. These sources help the forecasting system understand changes in security-related activities over time.

The knowledge gained from these areas provides the foundation for designing the proposed forecasting engine and integrating it with the MediRecon security framework.

2.2 Domain and Theoretical Foundations

2.2.1 Cybersecurity in IoMT Healthcare Systems

The Internet of Medical Things (IoMT) refers to medical devices and healthcare systems connected through communication networks. Examples include patient monitoring devices, medical imaging systems, hospital workstations, and healthcare servers.

IoMT technologies improve healthcare services by allowing medical information to be shared efficiently between connected systems. However, connecting multiple medical devices also increases the number of possible cybersecurity threats.

Healthcare systems may face unauthorized access, malware attacks, network intrusions, and data breaches. These incidents can affect patient information, hospital operations, and the availability of important medical services.

Many healthcare environments also contain devices that use older software or require special maintenance procedures. This can make security management more challenging.

Therefore, continuous security monitoring is important for identifying unusual activities and supporting the protection of healthcare systems.

The proposed Component 3 supports this requirement by analyzing security information collected from different sources and predicting possible future cybersecurity risk levels.

2.2.2 Fileless Malware and Behavioral Monitoring

Fileless malware is a type of malicious activity that uses legitimate system tools and processes to perform unauthorized operations. Unlike traditional malware, fileless attacks may not require a separate malicious executable file.

Attackers can misuse tools such as PowerShell, Windows Management Instrumentation (WMI), and other Windows administration utilities. Since these tools are also used for normal system management, identifying suspicious activities can be challenging.

Traditional antivirus systems often depend on known malware signatures. However, detecting fileless attacks may require monitoring system behavior, command execution, and network communications.

Behavioral monitoring examines activities such as unusual PowerShell commands, unexpected process execution, suspicious network connections, and authentication attempts.

For the proposed forecasting engine, these activities provide useful security indicators. Instead of analyzing only one suspicious event, the system considers changes in security-related activities across multiple time periods.

This approach helps the forecasting engine identify patterns that may be associated with increasing cybersecurity risk.

2.2.3 Security Telemetry and Time-Series Data

Security telemetry refers to information collected from computers, networks, applications, and security monitoring tools.

Examples include network connections, PowerShell execution events, process creation, authentication attempts, and DICOM communication activities.

These observations help security systems understand what is happening within the monitored environment.

Time-series data is information collected in chronological order. In cybersecurity, this allows security activities to be examined over a period of time rather than individually.

For example, an increase in network connections or suspicious PowerShell activities over several observation periods may provide useful information about changes in security conditions.

The proposed Component 3 organizes collected security information into five-minute observation windows. Each window contains 28 security-related features.

The system then uses twelve consecutive windows, representing one hour of security activity, to support the prediction of cybersecurity risk for the next 24 hours.

This method allows the forecasting engine to consider previous security behavior when producing a prediction.

2.2.4 Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) is a deep-learning technique designed to analyze information collected in sequences.

Unlike traditional machine-learning methods that may process individual observations independently, LSTM networks can learn relationships between current and previous observations.

This makes LSTM suitable for applications where information changes over time, such as financial forecasting, weather prediction, and cybersecurity monitoring.

In cybersecurity, suspicious activities may develop gradually. For example, unusual network connections, repeated command execution, and changes in endpoint activity may occur across different time periods.

An LSTM model can examine these sequential patterns and use the learned relationships to produce predictions.

In the proposed forecasting engine, the LSTM model receives security information collected during the previous one-hour period.

Based on this information, the model predicts the cybersecurity risk level for the next 24 hours.

The prediction is presented using four categories:

- LOW: A lower predicted level of cybersecurity risk.

- MEDIUM: A moderate predicted level of cybersecurity risk.

- HIGH: An elevated predicted level requiring security attention.

- CRITICAL: A severe predicted level requiring immediate investigation.

These categories help SOC analysts understand the predicted security condition and prioritize their monitoring activities.

2.3 Related Research

2.3.1 Machine Learning in Cybersecurity

Machine learning has become an important technology in cybersecurity because it can identify patterns within large amounts of security information.

Machine-learning techniques are commonly used for malware detection, network intrusion detection, spam filtering, and abnormal activity identification.

These techniques learn from data and use the identified patterns to classify new observations.

For example, a network intrusion detection system may analyze network traffic characteristics to identify unusual communication behavior.

However, many cybersecurity machine-learning applications focus on detecting existing attacks rather than predicting future risk.

The proposed Component 3 extends the monitoring approach by focusing on future cybersecurity risk prediction using previously collected security activities.

2.3.2 Deep Learning for Security Prediction

Deep learning is a branch of machine learning that uses neural networks to learn complex patterns from data.

Deep-learning methods can be useful when security information contains relationships that are difficult to identify using simple rules.

In cybersecurity, deep learning can support the analysis of network traffic, malware behavior, and endpoint activities.

LSTM networks are particularly relevant when security information is collected over time because they can examine the order of observations.

For example, a series of unusual endpoint activities followed by changes in network communication may provide useful information about the security condition of a system.

The proposed forecasting engine uses LSTM to examine sequential security telemetry and predict future risk levels.

This supports the research objective of moving beyond individual security alerts toward time-based security prediction.

2.3.3 Cybersecurity Monitoring in Healthcare Environments

Healthcare cybersecurity requires special attention because medical systems support important clinical operations.

Hospitals commonly use connected workstations, medical imaging servers, and network-based healthcare applications.

Cybersecurity incidents affecting these systems may interrupt medical services or expose sensitive information.

Network monitoring tools can identify unusual communications, while endpoint monitoring systems can detect suspicious processes and command execution.

However, monitoring information may be distributed across different tools, making it difficult to obtain a complete understanding of security conditions.

A centralized security monitoring approach can help combine information from different sources.

The proposed Component 3 uses network, endpoint, authentication, and DICOM-related telemetry to support future cybersecurity risk prediction within the MediRecon framework.

2.3.4 Comparison of Existing Approaches

Different cybersecurity technologies perform different functions.

Approach Main purpose Limitation in relation to forecasting

Traditional antivirus Detect known malware Mainly focuses on known malicious patterns

Network intrusion detection Identify suspicious network traffic Usually detects current network activity

Endpoint detection Monitor suspicious host behavior Mainly focuses on observed endpoint events

SIEM systems Collect and correlate security events Future-risk prediction is not inherent to rule-based correlation

Machine-learning detection Identify patterns associated with attacks Often focuses on current-event classification

LSTM-based forecasting Analyze previous activities to predict future conditions Requires properly organized sequential data

The comparison shows that traditional security monitoring and predictive security analysis serve different purposes.

The proposed forecasting engine does not replace existing detection mechanisms. Instead, it provides an additional capability that helps security analysts understand possible future cybersecurity risk.

2.4 Existing Systems and Techniques

2.4.1 Security Information and Event Management (SIEM)

Security Information and Event Management systems collect and analyze security events from different sources.

They provide centralized monitoring, alert generation, event correlation, and security reporting.

SIEM systems help SOC analysts investigate suspicious activities by presenting information collected from multiple devices and applications.

For example, a SIEM rule may generate an alert when several failed login attempts are detected within a short period.

However, traditional rule-based monitoring mainly focuses on identifying events that satisfy predefined conditions.

The proposed Component 3 provides an additional forecasting capability by analyzing previous security activities and estimating future cybersecurity risk.

2.4.2 Network and Endpoint Monitoring

Network monitoring examines communication between devices and systems.

It can provide information about network connections, protocols, communication destinations, and traffic activity.

Endpoint monitoring focuses on activities occurring within individual computers or workstations.

Examples include PowerShell execution, process creation, and authentication events.

Both monitoring approaches provide important security information. Network monitoring shows how systems communicate, while endpoint monitoring provides information about local system behavior.

The proposed forecasting engine combines these types of observations into a structured set of telemetry features.

This allows the LSTM model to examine changes in security-related activities over time.

2.4.3 Predictive Security Monitoring

Predictive security monitoring uses previously collected information to estimate possible future cybersecurity conditions.

Unlike traditional detection, which focuses on identifying existing suspicious activity, predictive monitoring attempts to provide early awareness of increasing security risk.

This can support SOC teams by helping them prioritize investigations and prepare appropriate responses.

In Component 3, the forecasting model produces a next-24-hour risk classification.

The result is displayed through a security dashboard containing a risk gauge, forecast history, telemetry information, and recommended security responses.

When a HIGH or CRITICAL risk level is predicted, the system also supports automatic email notification to the configured SOC/Admin recipient with an attached security report.

This connects predictive analysis with practical security monitoring and reporting activities.

2.4.4 Security Event Correlation

Security event correlation involves examining relationships between different security observations.

For example, suspicious PowerShell execution followed by unusual internal network communication may require further investigation.

Analyzing related observations can help security analysts understand possible connections between endpoint and network activities.

The proposed Component 3 includes a supporting potential threat-propagation feature that examines relevant endpoint and network evidence.

The dashboard presents possible relationships between monitored devices and network destinations.

However, a potential threat relationship does not automatically confirm that malware has spread between devices. Further investigation is necessary before identifying an actual security incident.

This feature provides additional context for SOC analysts alongside the future-risk prediction.

2.5 Research Gap and Summary

The reviewed concepts show that existing cybersecurity technologies provide several important capabilities, including malware detection, network monitoring, endpoint analysis, and security event correlation.

However, these technologies commonly focus on activities that have already occurred or are currently being observed.

Future cybersecurity risk prediction requires an additional approach that considers how security-related activities change over time.

The research gap addressed by Component 3 is the need for a forecasting mechanism that can analyze different types of IoMT security telemetry and provide an understandable prediction of the cybersecurity risk level for the next 24 hours.

The proposed Predictive IoMT Threat Forecasting Engine addresses this requirement through the following capabilities:

1. Collecting security information from network, endpoint, authentication, PowerShell, and DICOM-related sources.

2. Organizing telemetry into five-minute observation windows.

3. Using an LSTM model to analyze one hour of previous security activity.

4. Predicting the next-24-hour cybersecurity risk as LOW, MEDIUM, HIGH, or CRITICAL.

5. Presenting predictions through an interactive dashboard with historical trends and security indicators.

6. Supporting SOC investigation through security reporting, automated notifications, and potential threat-propagation information.

These capabilities allow Component 3 to complement the other MediRecon components by introducing a future-oriented security monitoring function.

The literature reviewed in this chapter provides the theoretical foundation for the research methodology and system architecture discussed in the following chapters.

3 RESEARCH METHODOLOGY

Chapter focus. A design-and-evaluate approach guided the artifact: define operational requirements, implement the pipeline, test component contracts, integrate event sources, and inspect real dashboard outputs and email/report delivery. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

3.1 Research design

A design-and-evaluate approach guided the artifact: define operational requirements, implement the pipeline, test component contracts, integrate event sources, and inspect real dashboard outputs and email/report delivery.

3.2 Data and study materials

The model development workflow used a threat-prediction dataset obtained for research. The deployed system consumes telemetry from authorized lab devices. Dataset publisher, licence, sampling protocol, and partition records must be checked against original project evidence before final submission.

3.3 Experimental procedure

Event streams are normalized into 28 named features at five-minute resolution. The prediction input is a chronological sequence of twelve windows. Inference results are stored with timestamps and surfaced through an API and dashboard.

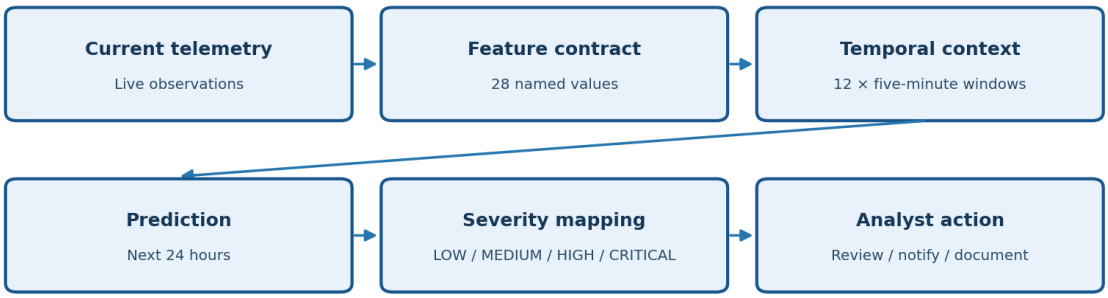


Figure 3.0.1 MediRecon overall research component architecture

Telemetry family	Representative features
Context	hour, is_off_hours
DICOM	C-STORE, C-FIND, C-MOVE, failed associations, bytes, external destinations
Network	flows, unique destinations, SYN, RST, SMB, WinRM, WMI/RPC, outbound, entropy
Endpoint	PowerShell counts, encoded, IEX, DownloadString, suspicious keywords, WMI, PsExec, process creation, suspicious parents, remote and failed logons

Table 1. Telemetry Feature Families

3.4 Measures and evaluation

Technical verification includes API availability, schema compatibility, successful model loading, sequence readiness, event ingestion, forecast persistence, and email attachment delivery. Predictive accuracy metrics should be reported only when backed by an archived held-out evaluation.

3.5 Data analysis

LOW, MEDIUM, HIGH, and CRITICAL are ordered risk categories for decision support. Probability distributions describe the model output for a particular inference, not a validated accuracy estimate. Persistent records support time-series visualization.

3.6 Ethics and privacy

The testbed uses authorized devices and telemetry. Reports should minimize personal information, control access to logs, and avoid representing a correlated destination as infected. SMTP credentials are configured outside source code.

3.7 Reproducibility

Reproducibility depends on versioned model and scaler files, feature ordering, preprocessing, sequence length, window duration, API contracts, and test records. The thesis documents observable implementation characteristics without inventing unseen experimental outcomes.

4 PRODUCT AND SOLUTION DESIGN

Chapter focus. Functional requirements include event ingestion, device discovery, 28-feature aggregation, LSTM inference, forecast history, live status, SOC notification, and PDF reporting. Non-functional requirements include traceability, schema stability, minimal device hardcoding, and safe handling of credentials. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

4.1 Requirements

Functional requirements include event ingestion, device discovery, 28-feature aggregation, LSTM inference, forecast history, live status, SOC notification, and PDF reporting. Non-functional requirements include traceability, schema stability, minimal device hardcoding, and safe handling of credentials.

Requirement	Implementation contract
Window interval	5 minutes
Sequence length	12 windows
Feature dimensions	28
Forecast horizon	Next 24 hours
Risk categories	LOW / MEDIUM / HIGH / CRITICAL
API port	8003 in documented lab

Table 2. Functional Requirements and Implementation

4.2 System architecture

Collectors operate on monitored devices and forward network, endpoint, PowerShell, process, and authentication evidence to the central FastAPI service. C1 provides the DICOM-oriented interface. Aggregation builds five-minute feature windows and the LSTM consumes twelve ordered windows.

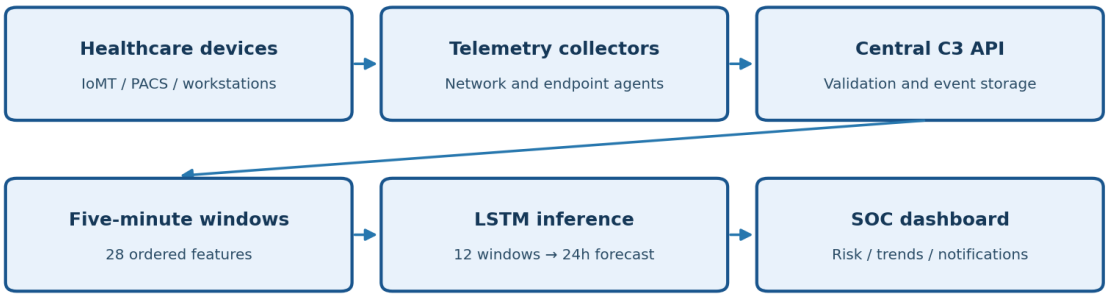


Figure 4.0.1 Overall Component 3 layered architecture

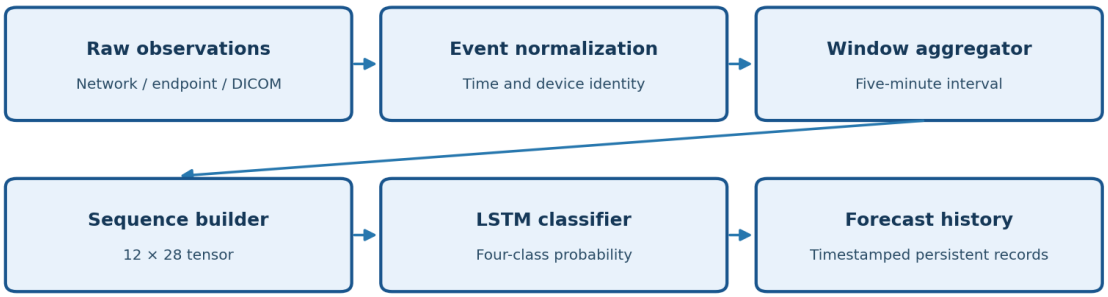


Figure 4.0.2 End-to-end telemetry and forecasting data flow

4.3 Design decisions

The five-minute cadence is part of the trained model's temporal contract. Changing it to two minutes without a compatible temporal redesign would alter the effective history duration. Dynamic device identifiers permit new hosts without editing the prediction model.

4.4 Detailed design

Two related but distinct paths are maintained: the prediction plane processes numeric time windows, while the evidence plane stores timestamped events and correlates potentially suspicious endpoint activity with internal network relationships.

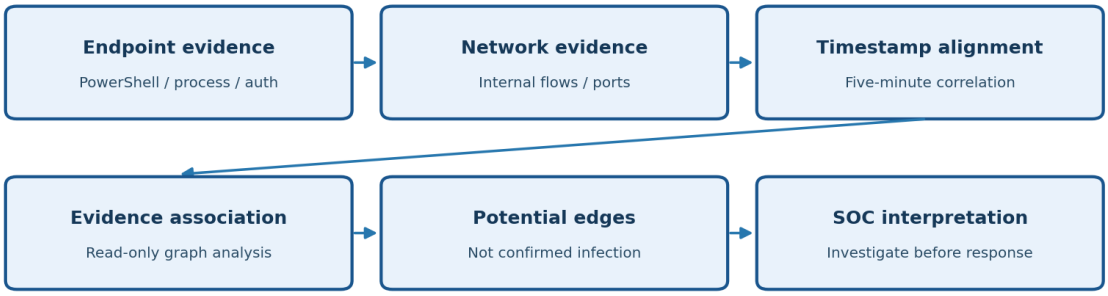


Figure 4.0.3 Evidence-based potential threat-spread correlation

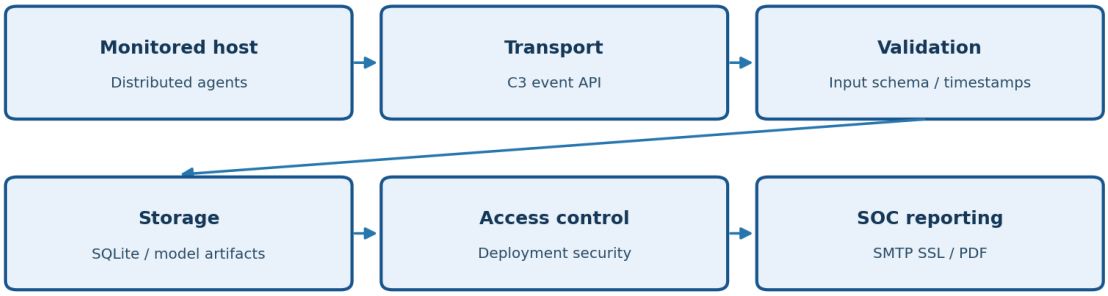


Figure 4.0.4 Component security and trust boundaries

4.5 Standards and practice

The design adopts SOC-oriented evidence preservation, timestamps, controlled APIs, role-appropriate reporting, and risk language. Potential propagation denotes correlated observations, not verified compromise or confirmed lateral movement.

5 IMPLEMENTATION INTEGRATION AND TESTING

Chapter focus. The implementation uses Python, FastAPI for central ingestion and forecasting endpoints, a dashboard presentation layer, SQLite for persistent history, a serialized LSTM and preprocessing scaler, and SMTP SSL for email notification. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

5.1 Technology stack

The implementation uses Python, FastAPI for central ingestion and forecasting endpoints, a dashboard presentation layer, SQLite for persistent history, a serialized LSTM and preprocessing scaler, and SMTP SSL for email notification.

5.2 Component implementation

Endpoint monitoring records relevant Windows process and security events. A distributed network agent aggregates network observations before forwarding them. The server constructs the fixed feature schema and runs the forecasting workflow.

5.3 Integration interfaces

DICOM telemetry is represented within the existing C3 feature contract and event store. Cross-layer evidence is available to correlation workflows. Other components retain responsibility for their own detection and incident-analysis logic.

5.4 Verification

Screenshots demonstrate a live service, twelve-window readiness, LOW and CRITICAL displays, observed indicators, persistent forecast trend, and a delivered CRITICAL notification with PDF attachment. These are functional observations, not a claim of population-level detection accuracy.

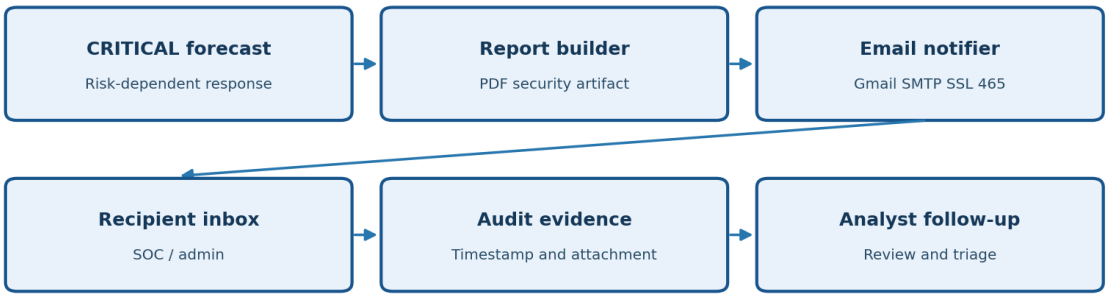


Figure 5.0.1. SOC reporting and email delivery workflow

Test observation	Evidence / interpretation
Live API and status	Dashboard shows LIVE status
History readiness	12 / 12 sequence windows shown
Forecast categories	LOW and CRITICAL screen captures
SOC notification	Successful send status and inbox receipt
PDF attachment	Report file visible in received email

Table 3. Verification Observations

5.5 Deployment

The central API listens on port 8003 in the documented lab deployment. A monitored endpoint runs the endpoint and network agents. The dashboard retrieves API state, and email configuration is provided through environment variables.

6 RESULTS AND VALIDATION

Chapter focus. The running system displays risk classes for the next 24 hours. The captured dashboard includes a LOW forecast and a CRITICAL forecast, each associated with the completed twelve-window input history. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

6.1 Research results

The running system displays risk classes for the next 24 hours. The captured dashboard includes a LOW forecast and a CRITICAL forecast, each associated with the completed twelve-window input history.

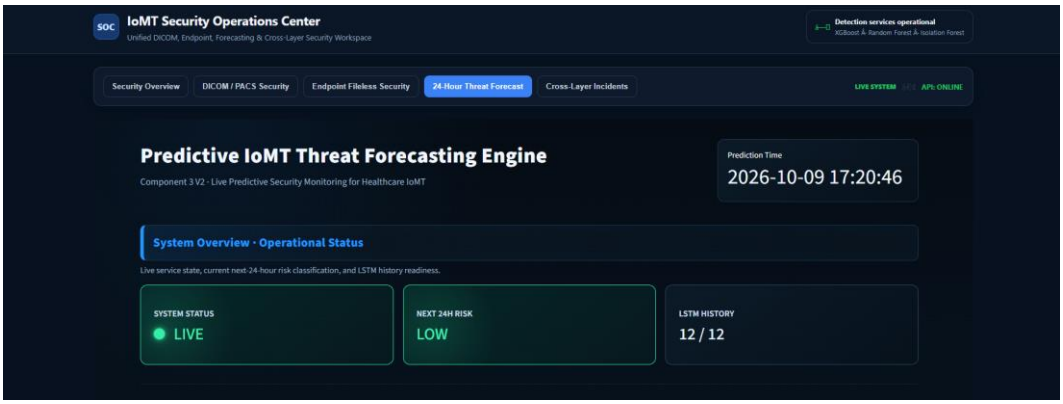


Figure 6.0.1 Supplied Component 3 operational screenshot

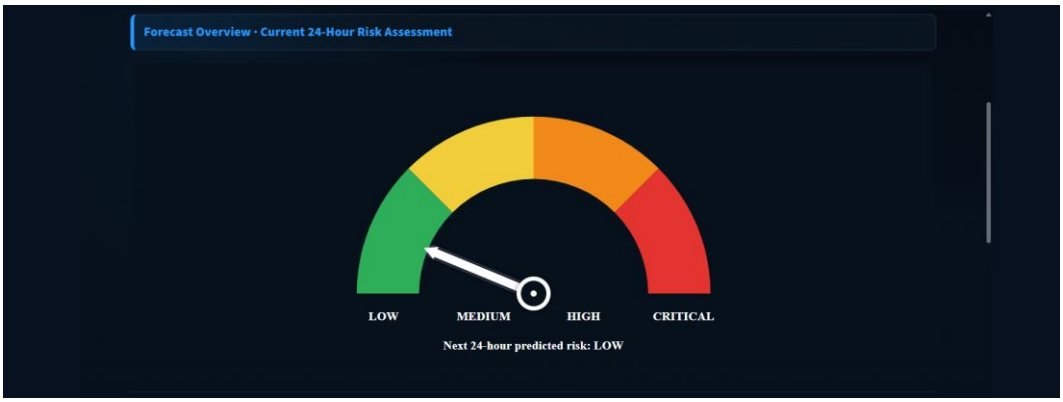


Figure 6.0.2 Supplied Component 3 operational screenshot

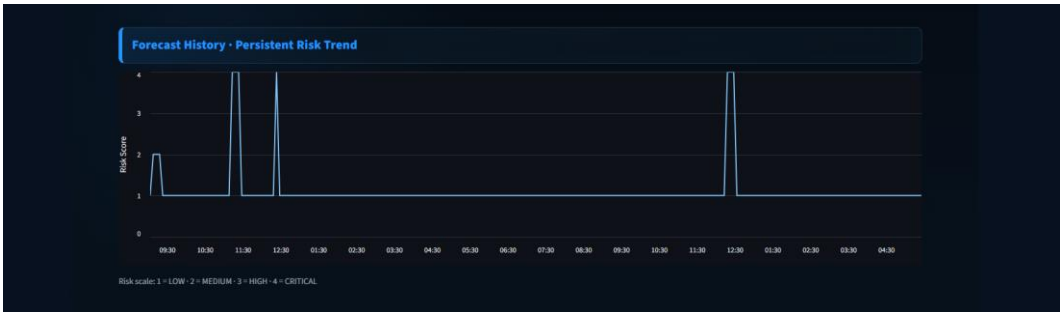


Figure 6.0.3 Supplied Component 3 operational screenshot

6.2 Technical results

Screenshots show network, endpoint, DICOM, and five-minute window status indicators. The observed latest window includes 209 network flows, 11 PowerShell events, and 101 process events in one capture; these values are window-specific observations.



Figure 6.0.4 Supplied Component 3 operational screenshot

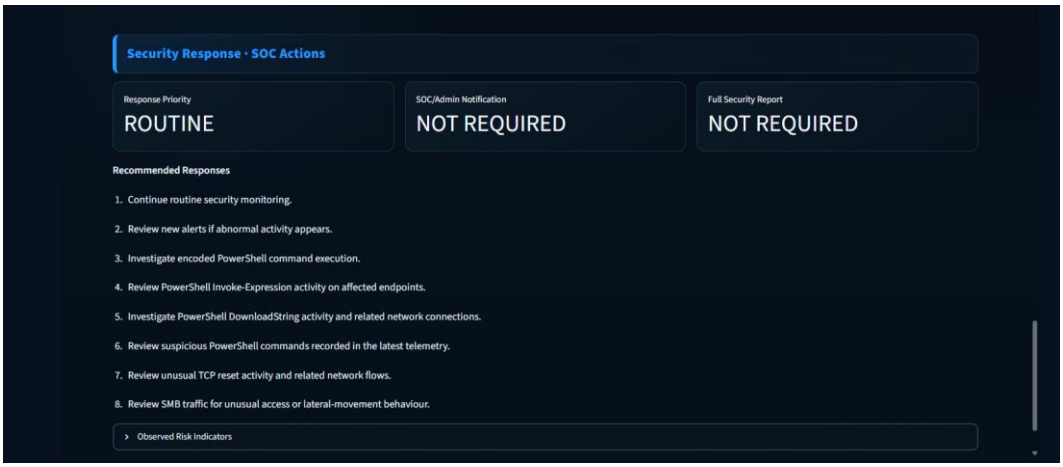


Figure 6.0.5 Supplied Component 3 operational screenshot

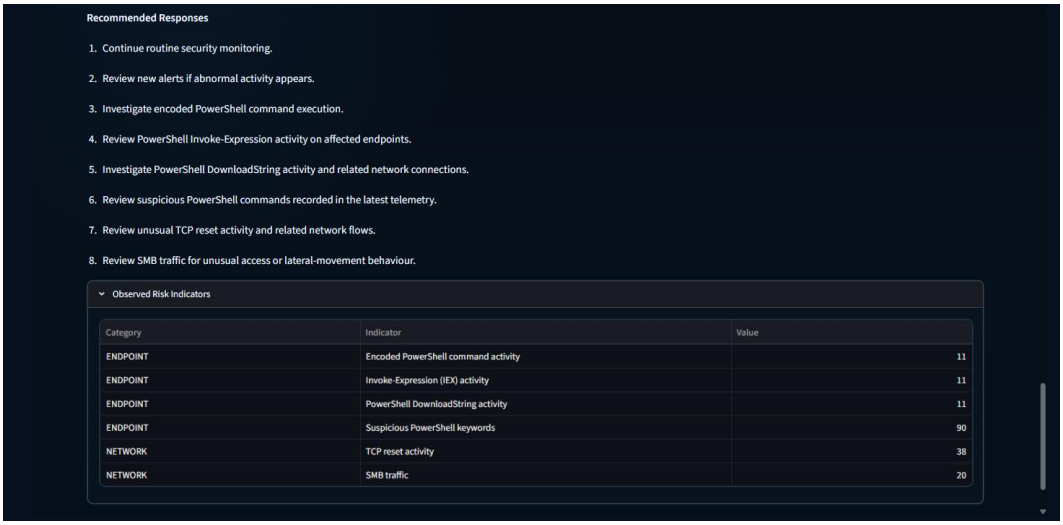


Figure 6.0.6. Supplied Component 3 operational screenshot

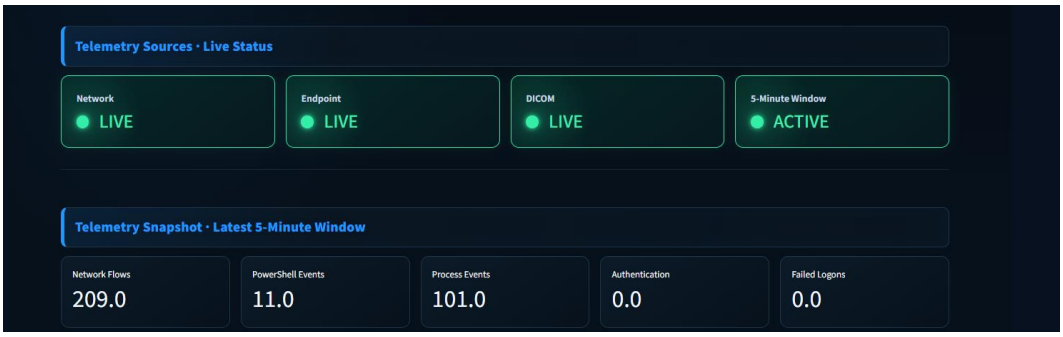


Figure 6.0.7 Supplied Component 3 operational screenshot

Observed latest-window feature	Displayed value
Network flows	209
PowerShell events	11
Process events	101
TCP resets	38
SMB flow count	20
Encoded-command count	11

Table 4. Latest Telemetry Window Observations

These values are transcribed from a single provided screenshot and must not be interpreted as averages, validated threat counts, or performance metrics.

6.3 Stakeholder validation

The interface provides current risk, a forecast gauge, history, risk indicators, recommended responses, and a reporting area. The display enables an analyst to distinguish risk forecast, source availability, and investigation evidence.

6.4 Integration results

A CRITICAL state was accompanied by a dashboard success message for automatic SOC email delivery. The supplied Outlook screenshots show receipt of a CRITICAL notification with a PDF attachment. This verifies that the email workflow functioned in the shown test.

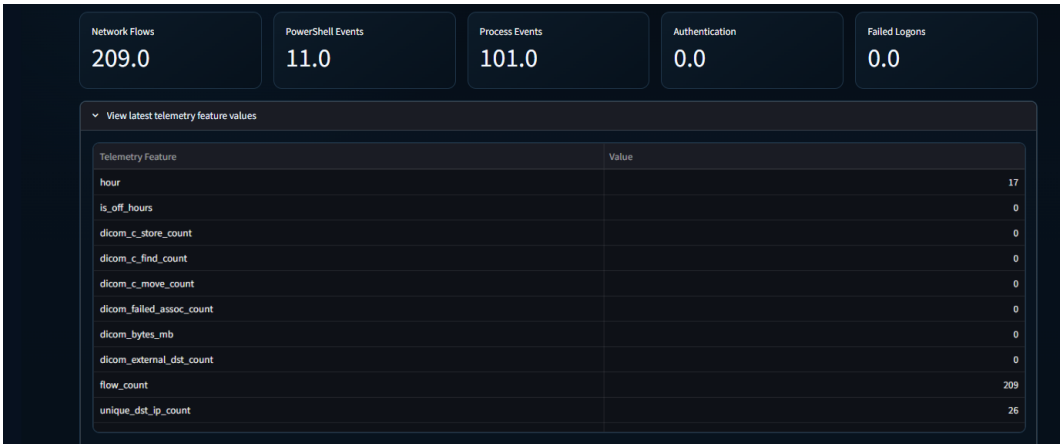


Figure 6.0.8 Supplied Component 3 operational screenshot

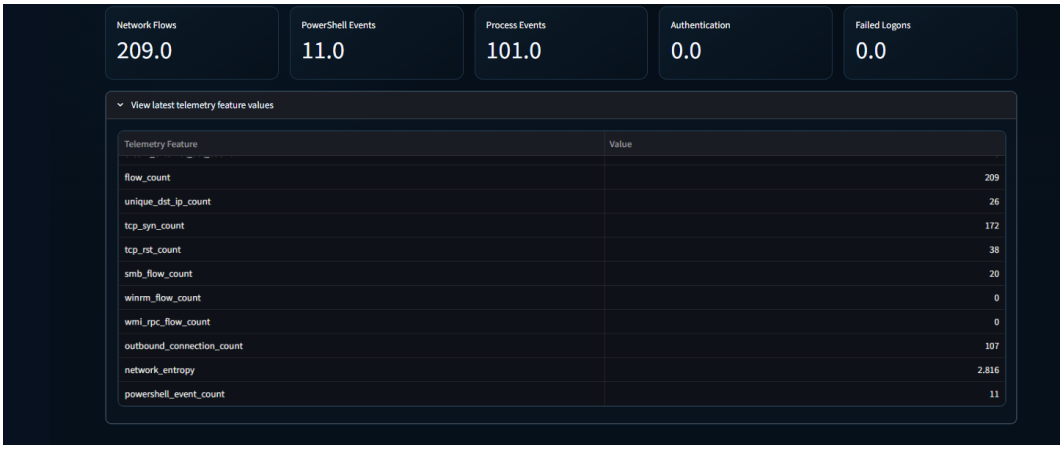


Figure 6.0.9 Supplied Component 3 operational screenshot

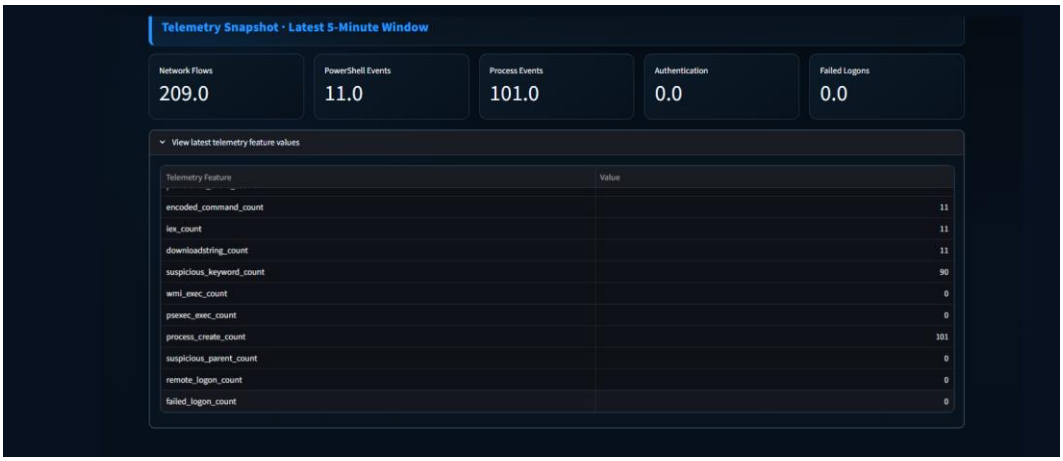


Figure 6.0.10 Supplied Component 3 operational screenshot

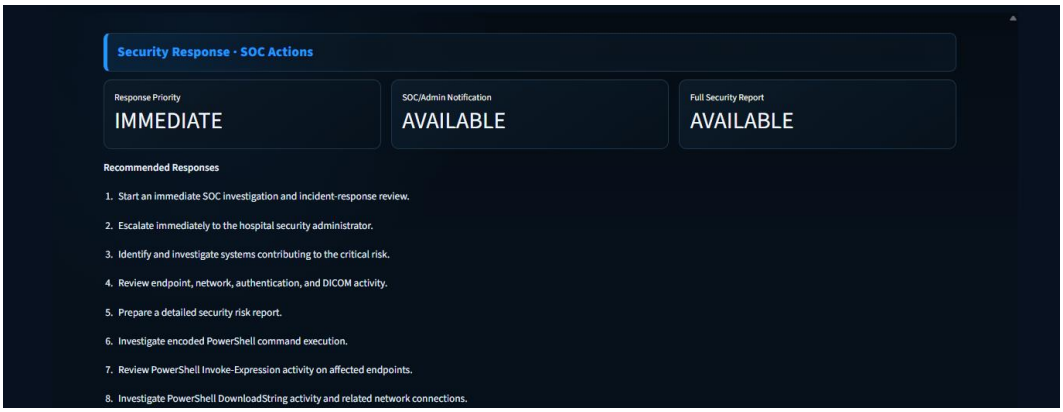


Figure 6.0.11 Supplied Component 3 operational screenshot

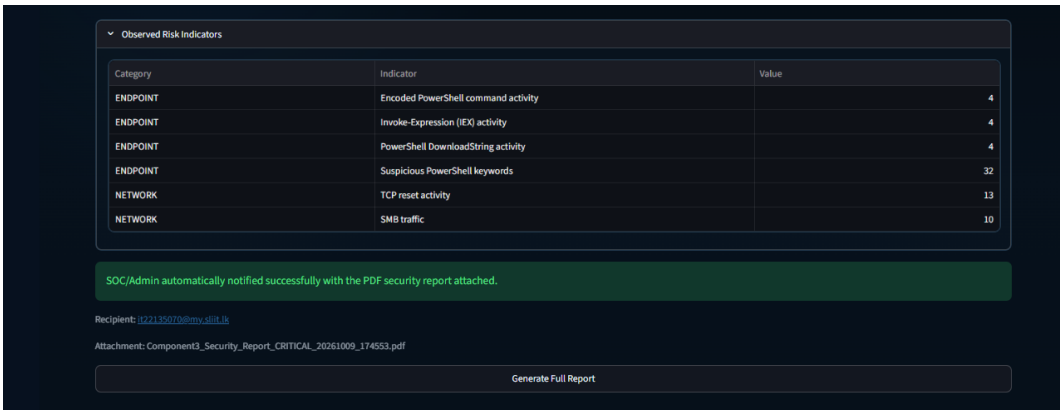


Figure 6.12 Supplied Component 3 operational screenshot

Notification verification note. The inbox capture and report attachment substantiate delivery for the illustrated test. Repeated messages may reflect repeated runs or notification triggers; the screenshot alone does not establish a deduplication guarantee.

6.5 Findings

The provided evidence supports functioning ingestion, display, risk categorization, historical plotting, and notification/report integration. Model generalization and comparative predictive performance require separately archived experimental results.

7 DISCUSSION

Chapter focus. A CRITICAL next-day forecast expresses a model-derived risk classification. It is not proof that an attack occurred, that malware executed, or that a destination was compromised. Analysts should combine forecasts with the underlying observed indicators. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

7.1 Interpretation

A CRITICAL next-day forecast expresses a model-derived risk classification. It is not proof that an attack occurred, that malware executed, or that a destination was

compromised. Analysts should combine forecasts with the underlying observed indicators.

7.2 Comparison with approaches

The forecasting engine differs from reactive rule-based SIEM logic by explicitly modeling time-ordered context. Its evidence plane retains direct event details so operational interpretation need not rely on a single categorical prediction.

7.3 Research contribution

The contribution lies in a reproducible sequence contract and its integration into an end-to-end security workflow: event collection, temporal aggregation, inference, persistence, visualization, and reporting.

7.4 Product contribution

The product integrates dynamic device monitoring and next-day risk forecasting with an evidence-aware SOC interface. It is designed as a component that can be connected to the broader MediRecon architecture.

7.5 Threats to validity

Results shown in screenshots are point-in-time system observations. Deployment conditions, event quality, class calibration, and differences between a lab network and clinical infrastructure affect interpretation; quantitative claims must be limited to verifiable test artifacts.

7.6 Implications

The component can help prioritize investigation and document risk signals. Human review remains essential before containment, credential revocation, or other potentially disruptive healthcare actions.

8 COMMERCIALISATION AND SUSTAINABILITY

Chapter focus. A forecasting layer may assist hospital SOC teams in prioritizing telemetry and documenting emerging risk across medical networks and endpoints. The

following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

8.1 Value proposition

A forecasting layer may assist hospital SOC teams in prioritizing telemetry and documenting emerging risk across medical networks and endpoints.

8.2 Market position

The concept is positioned as an additional analytical module for healthcare-focused security monitoring rather than a substitute for enterprise SIEM, EDR, or clinical device management.

8.3 Sustainability

Sustainable operation requires maintained collectors, secure storage, controlled notification policies, model version tracking, and staff training for interpreting predicted risk.

8.4 Scalability and compliance

The design supports dynamic device identities and distributed agents. Production adoption would require authentication, authorization, retention policy, resilience testing, and assessment against applicable healthcare data-protection obligations.

8.5 Sustainable development goals

Protecting healthcare service continuity supports the broader aims of good health and resilient digital infrastructure. The system's direct contribution is technical decision support, not a measured health-outcome intervention.

9 CONCLUSION AND FUTURE DIRECTIONS

Chapter focus. Component 3 implements a next-24-hour IoMT security-risk forecasting workflow that integrates telemetry collection, time-window aggregation, LSTM inference, evidence correlation, dashboard interpretation, and SOC reporting. The following sections document the individual research contribution, its design rationale, and its relationship to the operational system evidence.

9.1 Conclusion

Component 3 implements a next-24-hour IoMT security-risk forecasting workflow that integrates telemetry collection, time-window aggregation, LSTM inference, evidence correlation, dashboard interpretation, and SOC reporting.

9.2 Achievement of objectives

The supplied operational captures establish that the service displays live status, accepts multi-source telemetry, maintains sequence readiness, plots persistent risk history, and can deliver a CRITICAL email with a report attachment.

9.3 Individual contribution

The individual work brings together a fixed 28-feature temporal contract, twelve-step LSTM inference, persistent monitoring evidence, and a security-operations interface appropriate for component-level integration.

9.4 Future directions

Further work includes independently reproducible model evaluation, longitudinal hospital-like testing, secure multi-site deployment, stronger identity and access controls, and formal stakeholder usability assessment.

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GLOSSARY

Term	Meaning
IoMT	Internet of Medical Things
DICOM	Digital Imaging and Communications in Medicine
PACS	Picture Archiving and Communication System
SOC	Security Operations Center
LSTM	Long Short-Term Memory
SIEM	Security Information and Event Management
API	Application Programming Interface
SMTP	Simple Mail Transfer Protocol
C1/C2/C3/C4	MediRecon component identifiers

Table 5. Glossary Terms

APPENDIX A — FEATURE CONTRACT

Table 9.2. Component 3 Input Feature Contract

Index	Input feature
1	hour
2	is_off_hours
3	dicom_c_store_count
4	dicom_c_find_count
5	dicom_c_move_count
6	dicom_failed_assoc_count
7	dicom_bytes_mb
8	dicom_external_dst_count
9	flow_count
10	unique_dst_ip_count
11	tcp_syn_count
12	tcp_rst_count
13	smb_flow_count
14	winrm_flow_count
15	wmi_rpc_flow_count
16	outbound_connection_count
17	network_entropy
18	powershell_event_count

19	encoded_command_count
20	iex_count
21	downloadstring_count
22	suspicious_keyword_count
23	wmi_exec_count
24	psexec_exec_count
25	process_create_count
26	suspicious_parent_count
27	remote_logon_count
28	failed_logon_count

Table 6. Component 3 Input Feature Contract

APPENDIX B — API AND OPERATIONAL CHECKLIST

☐ Start central FastAPI service and verify model/scaler load.

☐ Confirm central API port and dashboard API base are consistent.

☐ Verify endpoint and network agents report a stable device identity.

☐ Check feature names, feature order and five-minute window timing.

☐ Wait for twelve chronological windows before treating the sequence as ready.

☐ Inspect persisted forecast history and risk-trend timestamps.

☐ Verify DICOM integration uses the existing 28-feature contract.

☐ Inspect evidence correlation without treating potential edges as confirmed infection.

☐ Verify SMTP configuration using environment variables without printing credentials.

☐ Test CRITICAL report creation and email delivery on an authorized system.

☐ Review generated report and remove any unnecessary identifying information.

APPENDIX C — ACADEMIC INTEGRITY AND EVIDENCE REGISTER

The author must verify student identity, original dataset citation and licence, model evaluation records, dates, version identifiers, and required AI-use disclosure before institutional submission. Screenshots are identified as supplied operational evidence; diagrams are explanatory architectural illustrations, not independent test results.